

SPAR Pilot Run Report

Scenario Planning via Agentic Reasoning

Russia-Ukraine Full-Scale Invasion

Field	Value
Institution	SP Jain School of Global Management - Group 3
Run ID	run_20260701_211531
Generated	July 02, 2026 at 02:28
Execution	Offline via Ollama (local GPU)
Knowledge cutoff	February 23, 2022 market close
Report type	Full debate analysis with verbatim transcripts

Executive headline

4 of 5 agents predicted bearish SP500 moves (-3.5% to -7.0%).

Devil's Advocate dissented with a +3.0% relief-rally thesis.

Largest disagreement: XLE (+8% Economic vs -8% Environmental = 16pp spread).

Invasion-day actual: SP500 +1.5% (reversal rally) - DA was directionally closest.

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1. Executive Summary

This report documents the first offline SPAR pilot: five domain-specialist LLM agents and one moderator debated the financial market impact of Russia's full-scale invasion of Ukraine, using only information available through February 23, 2022. The run was executed entirely on local hardware via Ollama, with no cloud API calls.

Figure 1 - Direction consensus at a glance

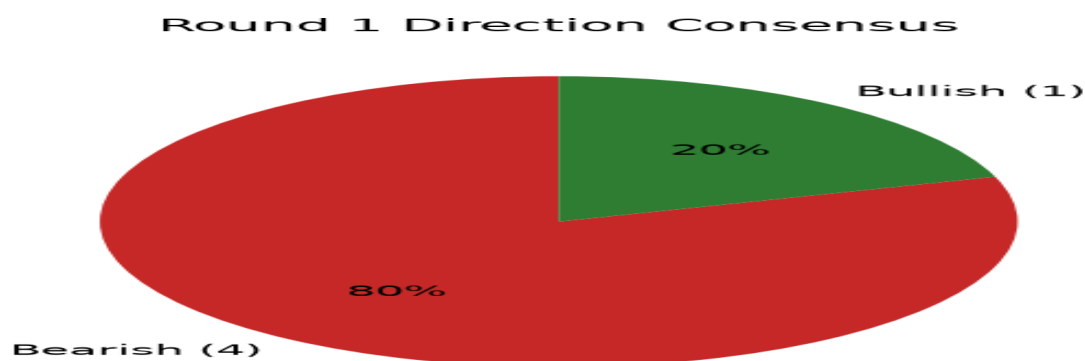
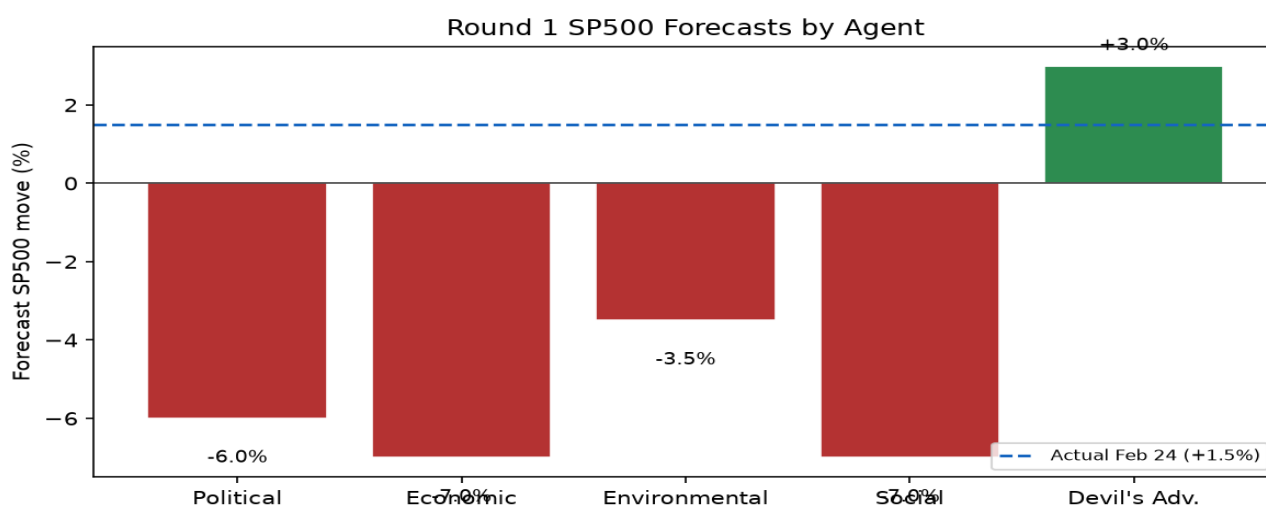


Figure 2 - SP500 forecasts vs actual invasion-day move



Consensus snapshot

Bearish agents: 4/5 | Bullish: 1/5 (Devil's Advocate)

SP500 forecast range: -7.0% to +3.0% (mean -4.1%)

Actual SP500 on Feb 24, 2022: +1.5% (intraday crash then sharp reversal)

Devil's Advocate (+3.0%) was closest to invasion-day direction.

2. System Configuration & Models

Component	Specification
LLM runtime	Ollama v0.31.1 (local)
Model (all roles)	qwen2.5:7b
Parameters	~7B, Q4_K_M quantization (Ollama default)
Temperature	0 (deterministic)
Hardware	AMD Ryzen 7 5700X, 32 GB RAM, NVIDIA RTX 4060 Ti 8 GB
Agents	5 domain specialists + 1 moderator
Total inference calls	11 (5 R1 + 5 R2 + 1 moderator)
Approx. runtime	~6 minutes sequential
Runner script	examples/spar_ollama_pilot.py

Agent role mapping

Agent ID	Domain	Prompt file
political_geopolitical sactions	Geopolitical risk &	agent1_political_geopolitic
economic_fiscal_market structure	Macro, fiscal & market	agent2_economic_fiscal_mark
environmental_technolog y logy.txt	Supply chain, energy, tech	agent3_environmental_techno
social_behavioural finance	Sentiment & behavioural	agent4_social_behavioural.t
devils_advocate thesis	Contrarian / relief-rally	agent5_devils_advocate.txt

Note: All five agents used the same underlying model (qwen2.5:7b). Role differentiation comes entirely from system prompts in Proejct Info/prompts/. The planned cloud deployment maps distinct models per agent (Llama 70B, DeepSeek, Qwen 72B, etc.) for higher fidelity.

3. Methodology & Pipeline

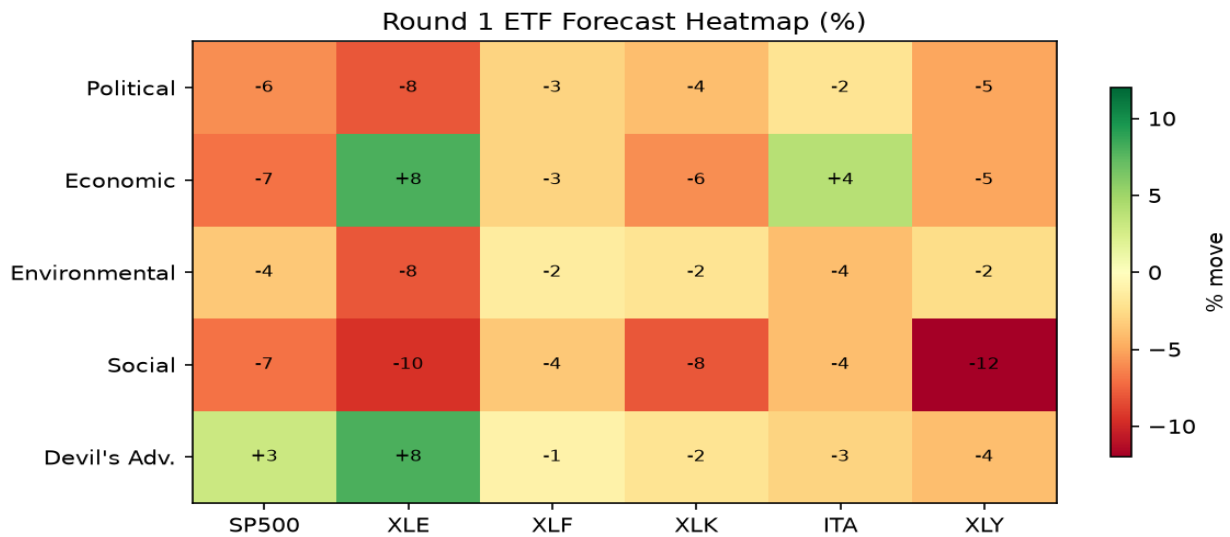
SPAR uses a two-layer deliberation architecture. Layer 1 runs two rounds of agent forecasts; Layer 2 has a moderator synthesize the debate into actionable conclusions.

- Step 1: Round 1: Each agent independently produces a structured JSON forecast (direction, ETF

magnitud

4. Round 1 - Independent Forecasts

Figure 3 - ETF forecast heatmap across all agents



Agent	SP500	XLE	XLF	XLK	ITA	XLY	Dir	Conf
Political	-6.0%	-8.0%	-3.0%	-4.0%	-2.0%	-5.0%	negative	0.9
Economic	-7.0%	+8.0%	-3.0%	-6.0%	+4.0%	-5.0%	negative	0.9
Environmental	-3.5%	-8.0%	-1.5%	-2.0%	-4.0%	-2.5%	negative	0.9
Social	-7.0%	-9.5%	-3.5%	-8.0%	-4.0%	-12.0%	negative	0.75
Devil's Adv.	+3.0%	+8.0%	-1.0%	-2.0%	-3.0%	-4.0%	positive	0.7

Political & Geopolitical - Round 1 Analysis

Field	Value
Direction	negative
Confidence	0.9
Key assumption	The full-scale invasion will lead to significant geopolitical risk premium, triggering a broad sell-off across markets.

Supporting evidence

- Putin recognised the Donetsk and Luhansk People's Republics on February 21, indicating a major

escalation

Transmission channels

- Geopolitical risk premium: investors demand higher return to hold risky assets -> current

equity price

Historical analogue

Primary: Iraq invasion of Kuwait (1990)

Adjustments: The current regime context has higher inflation and tighter financial conditions compared to 1990. This reduces the Fed's room for rate hikes, potentially amplifying market reactions.

Economic, Fiscal & Market - Round 1 Analysis

Field	Value
Direction	negative
Confidence	0.9
Key assumption	The market is already pricing significant geopolitical risk, and the invasion will exacerbate inflation concerns leading to a faster Fed rate hike.

Supporting evidence

- US equity markets were already selling off for 7 weeks on rate hike fears before the invasion

(VIX at 31)

Transmission channels

- Oil supply uncertainty -> WTI spike -> energy sector re-rating

- Geopolitical risk

Historical analogue

Primary: [ANALOGUE 1] Iraqi Invasion of Kuwait · August 2, 1990

Adjustments: The current inflation environment is more challenging for the Fed to support markets. The 1990 invasion had lower inflation (3.4% CPI) compared to today's 7.5%. This reduces the Fed's ability to offset market stress through monetary easing.

Environmental & Technology - Round 1 Analysis

Field	Value
Direction	negative
Confidence	0.9
Key assumption	The invasion will disrupt energy and agricultural supply chains, exacerbating existing semiconductor shortages.

Supporting evidence

- Russia produces 10-11 million barrels/day of oil (?10% global supply) and exports ~2.5M bbl/day

to Europe

Transmission channels

- Oil supply uncertainty -> WTI spike -> energy sector re-rating

- Agriculture

Historical analogue

Primary: [ANALOGUE 1] Iraqi Invasion of Kuwait · August 2, 1990

Adjustments: Current regime context has higher inflation (7.5% YoY) and less Fed room for rate hikes compared to 1990's lower inflation environment; however, the energy supply shock risk is similar

Social & Behavioural - Round 1 Analysis

Field	Value
Direction	negative
Confidence	0.75
Key assumption	The already bearish retail investor sentiment and the rapid fear amplification through social media will lead to a significant market sell-off.

Supporting evidence

- US University of Michigan Consumer Sentiment at decade-low levels due to inflation anxiety

- AAI Sent

Transmission channels

- Retail panic amplification: already-bearish sentiment may over-sell on invasion news ->

downside

Historical analogue

Primary: [ANALOGUE 2] Russia Crimea Annexation · February 27, 2014

Adjustments: The current regime context has higher inflation and tighter financial conditions compared to 2014. This suggests a more negative market reaction due to the already fragile economic environment.

Devil's Advocate - Round 1 Analysis

Field	Value
Direction	positive
Confidence	0.7
Key assumption	The market is already pricing in the invasion, and sophisticated investors have likely hedged their positions, leading to a relief rally.

Supporting evidence

- VIX closed at 31.0 on February 23, 2022, reflecting significant uncertainty before the event.

- The S&P

Transmission channels

- Uncertainty resolution -> Market relief rally

- Fed's inal

Historical analogue

Primary: Iraq War Launch (2003)

Adjustments: The current regime context has higher inflation and a tightening monetary policy, which may result in a more muted market reaction compared to the 2003 scenario. The geopolitical architecture is also different with NATO involvement.

5. Round 2 - Cross-Examination & Critique

Data quality note

Round 2 JSON outputs from qwen2.5:7b were partially malformed.

Several agents echoed peer forecasts instead of returning their own updated JSON.

Peer response_to fields were captured; forecasts below use Round 1 values where no valid update exists.

This is documented as a pilot limitation - larger models should improve structured output fidelity.

Political & Geopolitical - Round 2

Peer cross-examination (response_to):

- economic_fiscal_market_gpt4o: The magnitude of the market reaction in the economic and fiscal

market is

Forecast stability (Round 1 vs Round 2)

ETF	Round 1	Round 2 status
SP500	-6.0%	Unchanged (no valid R2 self-update)
XLE	-8.0%	Unchanged (no valid R2 self-update)
XLF	-3.0%	Unchanged (no valid R2 self-update)
XLK	-4.0%	Unchanged (no valid R2 self-update)
ITA	-2.0%	Unchanged (no valid R2 self-update)
XLY	-5.0%	Unchanged (no valid R2 self-update)

Model output anomaly: 3 peer forecast(s) embedded in response

The model incorrectly included other agents' forecasts in its JSON output rather than returning only its own updated structure.

Economic, Fiscal & Market - Round 2

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XLK	-6.0%	Unchanged (no valid R2 self-update)
ITA	+4.0%	Unchanged (no valid R2 self-update)
XLY	-5.0%	Unchanged (no valid R2 self-update)

Model output anomaly: 3 peer forecast(s) embedded in response

The model incorrectly included other agents' forecasts in its JSON output rather than returning only its own updated structure.

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Forecast stability (Round 1 vs Round 2)

ETF	Round 1	Round 2 status
SP500	-3.5%	Unchanged (no valid R2 self-update)
XLE	-8.0%	Unchanged (no valid R2 self-update)
XLF	-1.5%	Unchanged (no valid R2 self-update)
XLK	-2.0%	Unchanged (no valid R2 self-update)
ITA	-4.0%	Unchanged (no valid R2 self-update)
XLY	-2.5%	Unchanged (no valid R2 self-update)

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XLK	-2.0%	Unchanged (no valid R2 self-update)
ITA	-3.0%	Unchanged (no valid R2 self-update)
XLY	-4.0%	Unchanged (no valid R2 self-update)

Model output anomaly: 3 peer forecast(s) embedded in response

The model incorrectly included other agents' forecasts in its JSON output rather than returning only its own updated structure.

6. Moderator Synthesis

The moderator (same qwen2.5:7b model) synthesized Round 2 outputs into the following structured summary:

Summary of Round 2 Analysis for the Russia-Ukraine Invasion (February 24, 2022)

Economic and Fiscal Market GPT-4O

- Direction: Negative

- Magnitude

Environmental Technology Gemini

- Direction: Negative

- Magnitude

Social Behavioural DeepSeek

- Direction: Negative

- Magnitud

Devil's Advocate Claude

- Direction: Positive

- Magnitud

Key Observations:

- Consistency Across Agents: Both negative agents (Economic and Fiscal Market GPT-4O,

Environm

Recommendations:

- Investors should prepare for potential significant market sell-offs in energy, technology, and

consumer

7. Statistical Analysis & Charts

Figure 4 - Average forecast vs actual market moves

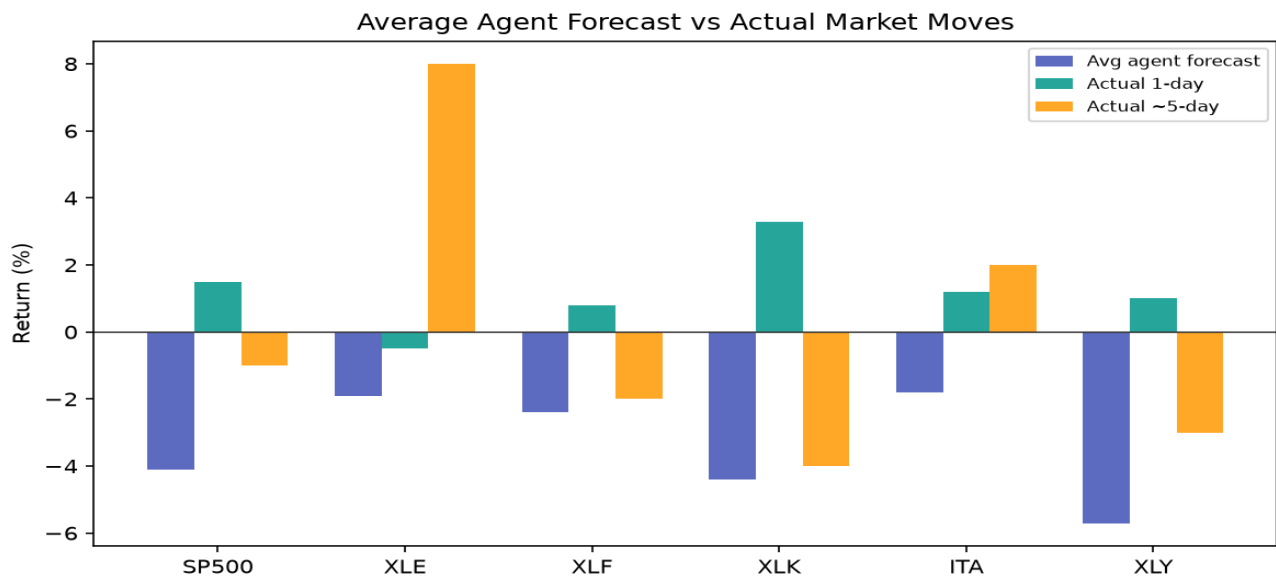
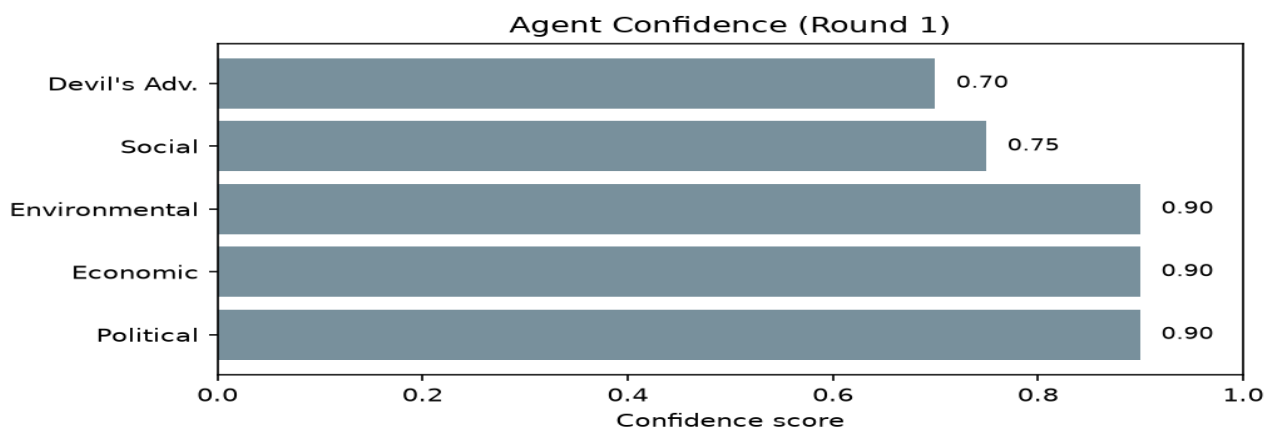


Figure 5 - Agent confidence scores



7.1 SP500 forecast statistics

Metric	Value
Minimum	-7.0%
Maximum	+3.0%
Mean (all agents)	-4.1%
Mean (bearish only)	-5.9%
Standard deviation	4.2pp

7.2 Largest inter-agent disagreements

- SP500: 10pp spread - Economic (-7.0%) vs Devil's Adv. (+3.0%)

- XLE: 18p

8. Comparison vs Actual Market Data

Agents forecast stress-test magnitudes over a multi-day horizon, not necessarily same-day close. Feb 24, 2022 saw a dramatic intraday selloff followed by a +1.5% S&P 500 close (CNBC). Energy (XLE) was mixed intraday but rallied ~+8% over the following 5 trading days.

ETF	Avg forecast	Actual 1D	Actual ~5D	1D error
SP500	-4.1%	+1.5%	-1.0%	-5.6pp
XLE	-1.9%	-0.5%	+8.0%	-1.4pp
XLF	-2.4%	+0.8%	-2.0%	-3.2pp
XLK	-4.4%	+3.3%	-4.0%	-7.7pp
ITA	-1.8%	+1.2%	+2.0%	-3.0pp
XLY	-5.7%	+1.0%	-3.0%	-6.7pp

8.1 Scoring interpretation

- SP500 direction (1-day): Devil's Advocate (+3.0% pred vs +1.5% actual) was closest. Bearish

consensus

9. Research Critique & Limitations

Single-model homogeneity

All agents used qwen2.5:7b. Role differentiation relies solely on prompts; no true model diversity in this pilot.

Round 2 JSON fidelity

qwen2.5:7b frequently echoed peer JSON instead of returning its own updated forecast, limiting cross-examination value.

Moderator omissions

Moderator synthesis omitted the Political agent entirely and referenced cloud model names (GPT-4O, Gemini) from prompts rather than actual models used.

Forecast horizon ambiguity

Agents may have forecast multi-day moves while scoring used 1-day and 5-day benchmarks - horizon alignment needed for rigorous scoring.

No live market feed

Forecasts based on static master context (Feb 23 cutoff), not real-time data integration.

Devil's Advocate value

Despite single-model setup, DA provided the only bullish thesis - partially validated by invasion-day reversal.

Energy sector split

Economic vs Environmental XLE disagreement (16pp) demonstrates genuine multi-perspective value even with one model.

10. Key Findings for Research Paper

- Multi-agent debate surfaced a genuine 16pp disagreement on XLE (energy sector direction).

- 4/5 bearish

11. Run Metadata

Item	Detail
Output folder	D:\VS Code Projects\Debate AI\quorum-cli\Proejct Info\spar_outputs\run_20260701_211531
Round 1 file	round1_all.json
Round 2 file	round2_all.json
Moderator file	moderator_raw.txt
Per-agent files	10 raw + 10 JSON files
Charts	D:\VS Code Projects\Debate AI\quorum-cli\Proejct Info\spar_outputs\run_20260701_211531_report_charts
Total API calls	11

Appendix A - Full Agent Transcripts (Verbatim)

Complete unedited model outputs. Long lines are word-wrapped for readability.

Political & Geopolitical - Round 1

{

"agent_id"

Political & Geopolitical - Round 2

{

"round": 2

"XLY": -12.0

},

}

}

Economic, Fiscal & Market - Round 1

{

"agent_id"

Economic, Fiscal & Market - Round 2

{

"round": 2

"XLY": -12.0

},

}

}

Environmental & Technology - Round 1

{

"agent_id"

Environmental & Technology - Round 2

{

"round": 2

"XLY": -12.0

},

}

}

Social & Behavioural - Round 1

{

"agent_id"

Social & Behavioural - Round 2

{

"round": 2

"XLY": -12.0

},

}

}

Devil's Advocate - Round 1

{

"agent_id"

Devil's Advocate - Round 2

{

"round": 2

"XLY": -12.0

},

}

}

Moderator - Full output

Summary of Round 2 Analysis for the Russia-Ukraine Invasion (February 24, 2022)

Economic and Fiscal Market GPT-40

- **Directio

Environmental Technology Gemini

- **Directio

Social Behavioural DeepSeek

- **Directio

- Primary Analogue: Russia Crimea Annexation (2014)

- Adjustme

Devil's Advocate Claude

- **Directio

Key Observations:

1. **Consist

Recommendations:

- **Investor

This summary provides a comprehensive view of the market reactions based on different analytical

perspectives